

Optimal Confidence Bounds for Sparse Random Projections

Sub-gamma calculus, negative association of moments, and Padé approximants

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Motivation

With high probability, **random linear maps are near-isometric**. A suitably sampled random $\Pi \in \mathbb{R}^{m \times n}$ with $m = O(\varepsilon^{-2} \log(1/\delta))$ preserves $\|x\|_2$ up to $1 \pm \varepsilon$ w.p. $1 - \delta$, per the **Johnson–Lindenstrauss guarantees** [1]. This is a cornerstone of high-dimensional statistics and algorithms, behind approximate nearest-neighbour search [2], [3], clustering [4], compressed sensing [5], linear-algebra sketching [6], [7], and LLM fine-tuning (VeRA) [8].

The catch: the hidden $O(\cdot)$ constant is **orders of magnitude** off from practice.

Concrete problem with sparse projections

[9]: best known constants differ from observed behaviour by $\approx 100 \times$.

This Paper’s Contribution

- Optimal concentration bound.** A sharp Bernstein-type bound meeting the exact leading constant 4, improving prior results by up to $\approx 90\%$ in regimes of practical interest.
- Novel analytical framework.** Negative Association of Moments (NAM); aggregation of conditional sub-gamma bounds via Padé approximants; Hanson–Wright-type bounds for randomly masked quadratic forms.
- Reproducible verification.** Python/Mathematica notebooks, CAD-based symbolic proofs of the key inequalities, and all figures/experiments at osf.io/5kr7t.

Setup: Sparsified Rademacher Projections

Sparsity trades density for speed. Each column keeps only s of m entries nonzero, so matrix-vector products cost $O(s)$ instead of $O(m)$ per column: $(\Pi)_{k,i} = s^{-1/2} \eta_{k,i} g_{k,i}$

- $g_{k,i}$ i.i.d. Rademacher (signs)
- $\eta_{k,i}$ picks s of m rows, without replacement, per column (mask)
- $s^{-1/2}$ keeps $\mathbb{E}[\|\Pi x\|_2^2] = \|x\|_2^2$

The object of study – the **distortion**: $Q \triangleq \|\Pi x\|^2 - \|x\|_2^2 = x^T (\Pi^T \Pi - I) x$.

Sparsity $s \in [1, m]$ trades matrix-vector product speed against concentration quality; $s = m$ recovers dense Rademacher projections.

Main Result

The tail is exactly Gaussian, at the sharp constant. No multiplicative loss in the exponent, and the scale term is fully explicit:

THEOREM 1

Let $v^2 = \text{Var}[Q] = \frac{2}{m} \sum_{i \neq j} x_i^2 x_j^2$ and $c = 3\rho/s$, $\rho = \|x\|_\infty / \|x\|_2 \leq 1$. For $|t| < 1/c$,

$$\mathbb{E}[e^{tQ}] \leq \exp\left(\frac{v^2 t^2}{2(1 - c|t|)}\right) \Rightarrow \mathbb{P}\{|Q| > \varepsilon\} \leq 2 \exp\left(\frac{-\varepsilon^2}{2(v^2 + c\varepsilon)}\right).$$

Solving for the dimension:

$$m \geq \frac{4 \log(2/\delta)}{\varepsilon^2} \cdot \frac{1}{1 - \frac{6\rho \log(2/\delta)}{s\varepsilon}}$$

Leading constant is exactly 4 – matching Burr et al. [10]’s sharp lower bound, not merely up to $O(\cdot)$.

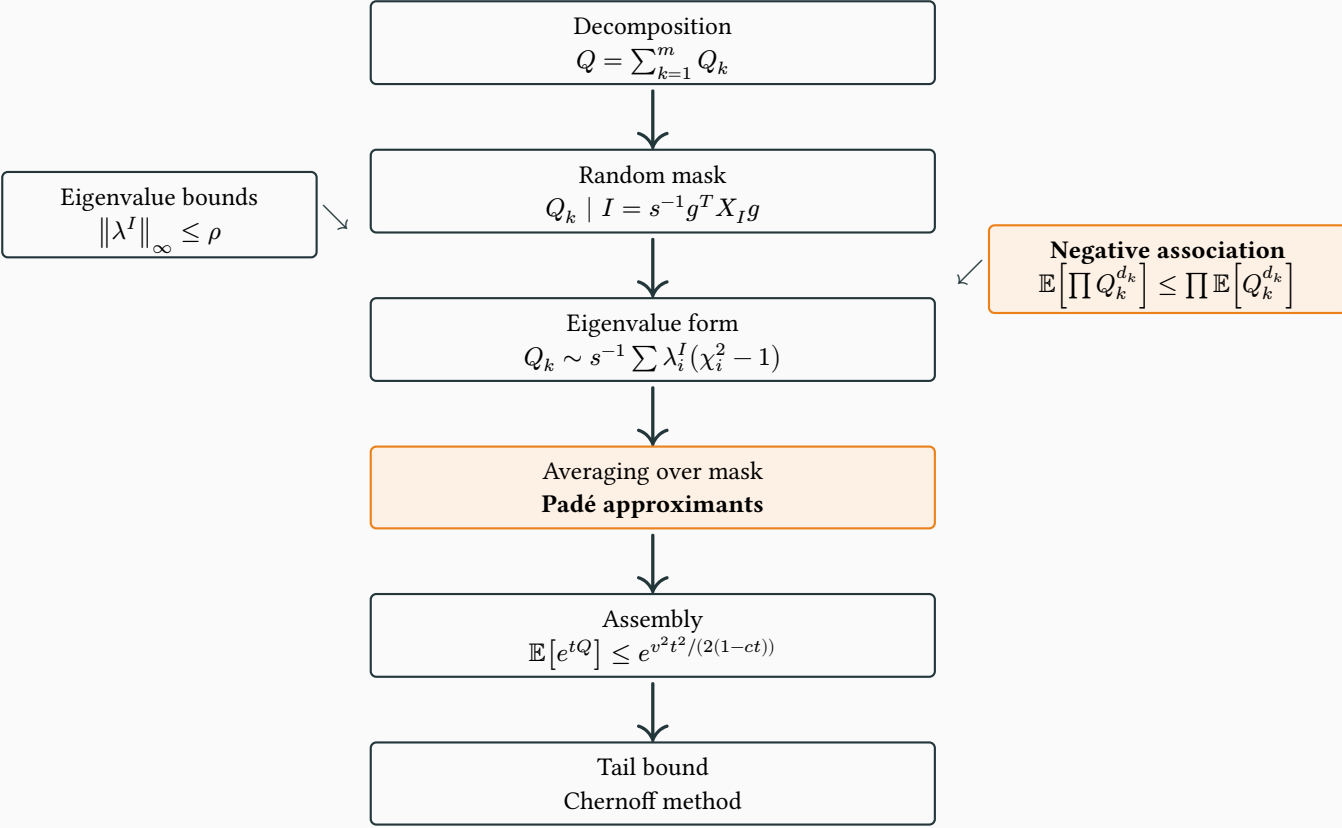
Where This Sits in the Literature

Prior work pays for sparsity in confidence. Every earlier sufficient bound below carries hidden constants that degrade δ exponentially; ours is the first to match the necessary bound exactly.

Reference	Bound on m	Technique
<i>Necessary</i>		
Kane–Meka–Nelson [11], Jayram–Woodruff [12]	$\Omega(\log(1/\delta)/\varepsilon^2)$	Comm. complexity
Burr et al. [10]	$4 \log(2/\delta)(1 + o(1))/\varepsilon^2$	Probabilistic arg.
<i>Sufficient (prior work)</i>		
Kane–Nelson ‘10 [13]	$\Omega\left(\max\left\{\frac{\log(1/\delta)}{\varepsilon^2}, \log^2 \frac{1/\delta}{\rho\varepsilon}\right\}\right)$	Chaos process
Kane–Nelson ‘12 [14]	$\Omega\left(\max\left\{\frac{\log(1/\delta)}{\varepsilon^2}, \frac{\log(1/\delta)}{\rho\varepsilon}\right\}\right)$	Moment bounds
Cohen ‘16 [15]	$\Omega\left(\max\left\{B \frac{\log(1/\delta)}{\varepsilon^2}, \log_B \frac{1/\delta}{\rho\varepsilon}\right\}\right)$	Matrix Chernoff
Cohen–Jayram–Nelson [16]	$\max\left\{128 \frac{\log(1/\delta)}{\varepsilon^2}, 8\sqrt{2} \frac{\log(2/\delta)}{\rho\varepsilon}\right\}$	Decoupling+MGF
<i>This work</i>		
Theorem 1	$\frac{4 \log(2/\delta)}{\varepsilon^2} \cdot \frac{1}{1 - \frac{6\rho \log(2/\delta)}{s\varepsilon}}$	Sub-gamma calc.

Proof Roadmap

Three moves turn a hard sum into a sharp tail.



(i) condition on the mask to expose a bounded spectrum; (ii) control mixed moments by negative association; (iii) assemble a sub-gamma MGF via rational approximants.

Tool 1: Negative Association of Moments (NAM)

Moments factorize even where NA fails. NAM is the weaker property Chernoff bounds actually need.

Definition (NAM)

$$X_1, \dots, X_m \text{ satisfy NAM if for all } \alpha_j \geq 0, \\ \mathbb{E}\left[\prod_j X_j^{\alpha_j}\right] \leq \prod_j \mathbb{E}[X_j^{\alpha_j}].$$

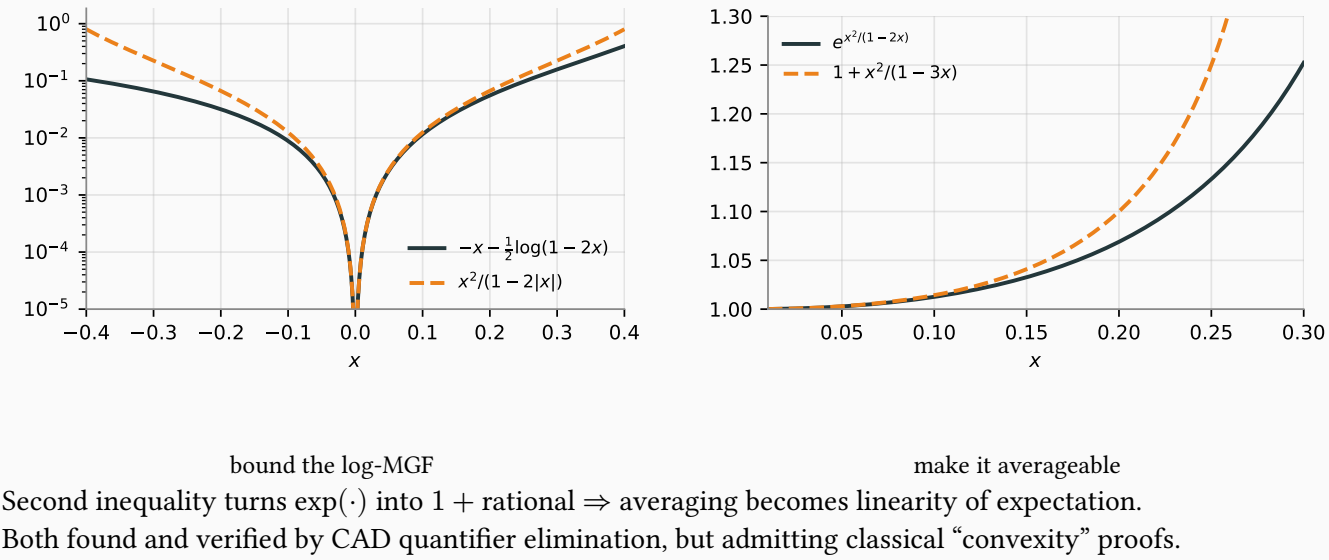
Equivalent to a **negative-association** condition on **mixed moments** – generalizes the notion behind Chernoff bounds under limited independence: $\mathbb{E}\left[\prod_{i \in S} X_i\right] \leq \prod_{i \in S} \mathbb{E}[X_i]$ for all S . The classical NA demands it for all *non-decreasing* f_j : a strictly stronger, different class [17].

NA genuinely fails here

For $n = 4$, $m = 3$, $s = 2$, $x \propto (1, \frac{3}{2}, 2, \frac{5}{2})$: exhaustive enumeration gives $\mathbb{E}[f(Q_1)f(Q_2)] > \mathbb{E}[f(Q_1)]\mathbb{E}[f(Q_2)]$ for $f = 1[\cdot > 0]$ – settling the open question of Dasgupta et al. [18].

Tool 2: Padé Approximants Instead of Taylor

Two nested rational bounds beat Taylor. Rational functions matching the Taylor series to prescribed order – sharper, over wider intervals, and *linear enough to average*. We nest two of them.



bound the log-MGF
Second inequality turns $\exp(\cdot)$ into $1 + \text{rational}$ $\Rightarrow$ averaging becomes linearity of expectation.
Both found and verified by CAD quantifier elimination, but admitting classical “convexity” proofs.

Tool 3: Aggregating Conditional Sub-Gamma Bounds

Averaging costs only the scale, not the variance.

THEOREM 3

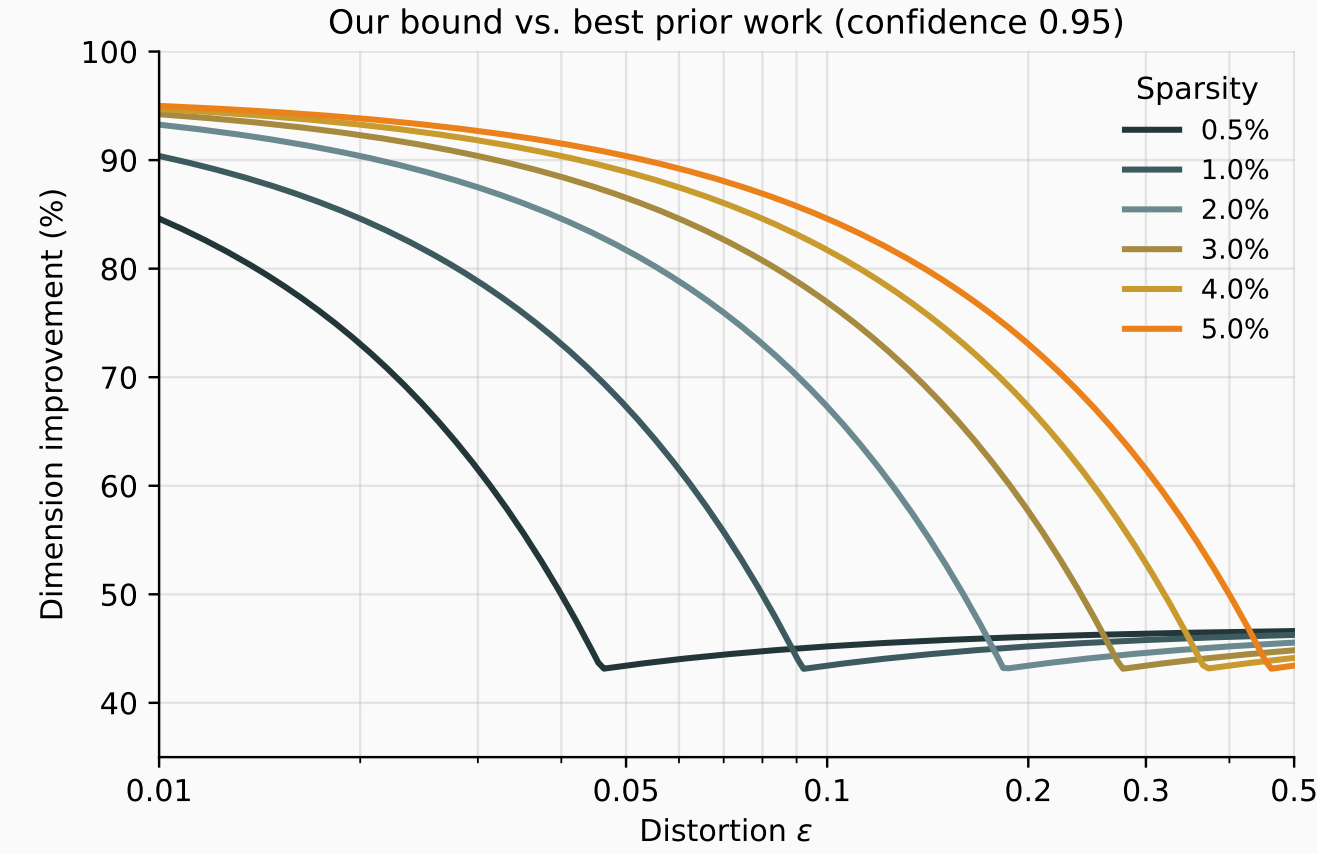
Let X, I auxiliary randomness with $\log \mathbb{E}[e^{tX} | I] \leq \frac{t^2 v_I^2}{2(1 - c_I t)}$, $\mathbb{E}[v_I^2] = \bar{v}^2$, $c_I \leq av_I$, $v_I \leq v_{\max}$ a.s. Then for $t < \frac{\sqrt{2}}{(a+1)v_{\max}}$,

$$\log \mathbb{E}[e^{tX}] \leq \frac{t^2 \bar{v}^2}{2(1 - (a+1)v_{\max} t / \sqrt{2})}.$$

Naive averaging loses the variance proxy; here $\bar{v}^2$ survives intact and only the *scale* degrades, by an explicit factor. Applied to our setting this recovers exactly $c = 3\rho/s$.

Quantitative Gain vs. Prior Bound

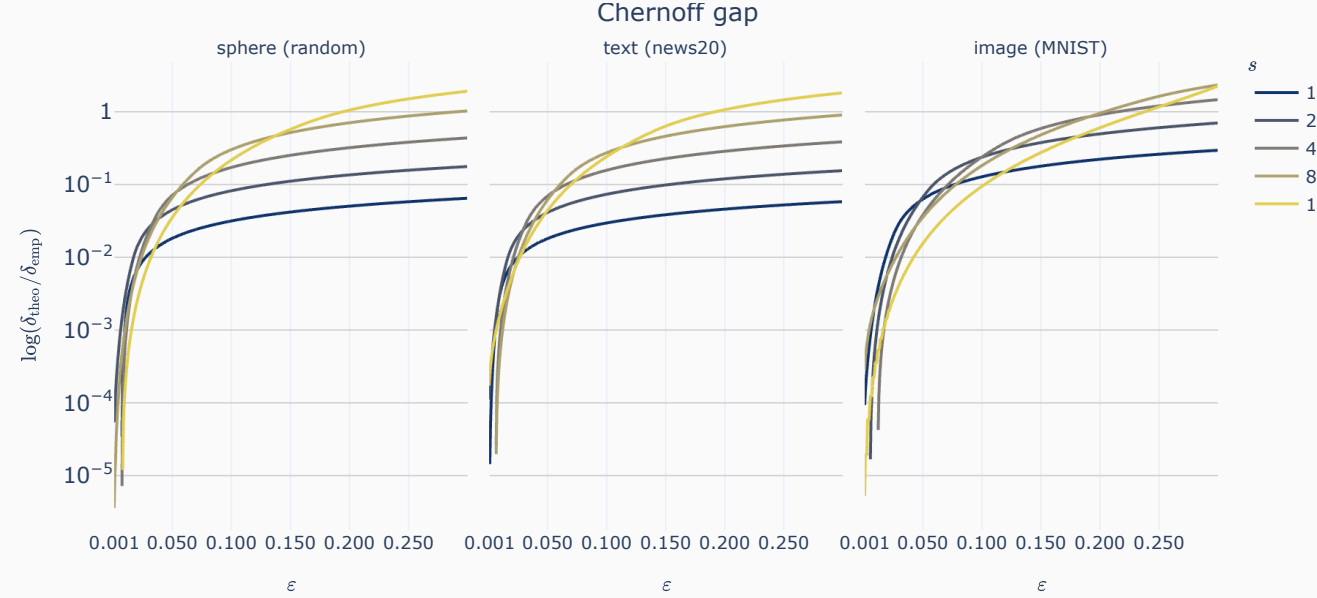
Same guarantee, a fraction of the dimension.



70–95% dimension reduction over the best prior data-oblivious bound [16] (confidence 0.95); gains grow with sparsity parameter s .

Empirical Check on Real Data

Theory tracks Monte-Carlo simulation almost exactly.



Empirical MGF $\rightarrow$ Chernoff $\rightarrow \delta_{\text{emp}}$ vs. Theorem 1’s δ_{theo} , on sphere / TF-IDF (news20) / MNIST inputs. **For $\varepsilon < 0.05$ the two agree to within a few percent**, gap decaying with ε .

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Takeaways & Future Work

- First sparse-JL bound with the **exact** leading constant 4**
- Matches Burr’s lower bound, not merely up to $O(\cdot)$**
- Reusable machinery: NAM, Padé nesting, conditional sub-gamma aggregation, masked Hanson–Wright**

- Results for structured matrices (circulant, Toeplitz) ?**
- Data-dependent variants exploiting input structure for adaptive sketching?**
- NAM beyond projections: other problems with moment sub-multiplicativity?**
- Constant-optimal general Hanson Wright inequality?**

Practical upshot. `scikit-learn` ships sparse random projections sized by earlier bounds — these results directly shrink the required dimension while strengthening the guarantee.

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Code, notebooks, CAD verification: osf.io/5kr7t